



KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

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T characteristics of the motor.

7. List and explain the merits of constant V/f control of Induction Motor.
8. Draw and explain the closed loop speed control scheme widely used.
9. A motor-drive system has the following specifications: Polar moment of inertia of motor-load system referred to the shaft, $J = 5 \text{ kg-m}^2$ Motor torque $T_m = 0.025N$ and Load Torque, $T_L = 0.025N$ where, “N” is the speed of the motor in rpm. Calculate the start-up time of the drive.
10. Explain the four-quadrant operation of a motor driving a passive load.
11. A 200 V, 20 A, 800 rpm, separately excited DC motor has an armature resistance of 0.5Ω . The motor drives a load whose torque-speed equation is $T_L = 0.025N$ where, “N” is the speed of the motor in rpm. The motor is driven from a single phase full bridge controlled rectifier, operating in continuous conduction mode, from a 230V, 50Hz AC source. Find the firing angle of the converter if the motor is running at 800 rpm.
12. What are the functions of power modulator in an electric drive?
13. A single phase fully controlled converter is used to control a DC separately excited motor of 200V, 900rpm, 100A with armature resistance of 0.06Ω . The supply is 210V, 50Hz. Determine firing angle for rated motor torque and 75% speed.
14. How speed of the induction motor can be controlled using stator frequency?
15. Draw the Torque – Speed characteristics of the following loads (i) Constant torque load (ii) Traction load
16. Draw the block diagram of a closed loop speed control of an electric drive.
17. Derive the speed torque characteristics of DC separately excited motor for constant field and speed torque characteristics for variable armature resistance.
18. Draw the circuit diagram of a class-C chopper fed DC motor. Draw its speed torque characteristics.
19. Differentiate between passive and active load torques. Give examples.
20. Draw and explain the speed torque curves of a fan load and traction load.
21. Why square wave inverter fed induction motor drives are not popular?
22. Explain the difference between the VSI fed induction motor drive and the CSI fed induction motor drive.

3. Explain the stator voltage control for Induction motor drive
4. Explain the working of v/f control of Induction motor drive
5. With the aid of a neat labelled circuit diagram, explain the operation of power-recovery scheme induction motor drive
6. Illustrate four quadrant operation of drive considering hoist as an example
7. Draw the armature voltage and armature current waveforms of 3 phase bridge rectifier fed DC motor drive for $\alpha=60^\circ$
8. What are the slip power recovery control schemes of induction motor drive? How is static Kramer drive is used to control the speed of induction motors?
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